PINDIKURU HARSHIL KUMAR

Associate Software Engineer - DevOps

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SUMMARY

DevOps engineer who built hands-on cost optimization judgment through manual cloud rightsizing work, then applied it to engineer an autonomous FinOps agent (Python, LangGraph) that encodes production-safety guardrails static tools ignore. Experienced across Azure automation, DevSecOps pipelines, and telemetry-driven infrastructure tooling, with a growing focus on AI observability for agentic systems.

TECHNICAL SKILLS

Cloud & Infra: Azure, AWS, Terraform, Bicep, Ansible, Kubernetes, Docker, Podman

DevSecOps & Security: Semgrep, Trivy, SonarQube, OWASP ZAP, Checkov, Gitleaks

Automation & Scripting: Python, Bash/Shell, KQL

AI & Observability: LangGraph, Langfuse, Azure Monitor, Prometheus, Grafana, Incident Response, RCA

CI/CD & OS: Azure DevOps, Jenkins, GitHub Actions, Git/GitHub, Linux (Ubuntu)

PROFESSIONAL EXPERIENCE

Associate Software Engineer - DevOps — Oct 2025 - Present

CoStrategix

Cost Optimization → FinOps Agent Architecture

- Manually audited live compute infrastructure and executed a targeted rightsizing initiative that reclaimed $13,400+ in annual cloud spend (40% reduction on audited resources), building hands-on judgment for safe, telemetry-informed rightsizing across Azure App and Function services.
- Applying that judgment, designed and built an autonomous multi-cloud FinOps agent (LangGraph, Python; Azure, Databricks, Snowflake) via a ProviderRegistry abstraction that automates rightsizing analysis while encoding production-safety guardrails a human analyst applies instinctively but static tools ignore.
- Engineered a KQL exception-log audit step as an SRE crash-detection guardrail, blocking downsizing recommendations on services in active crash loops before they reach a human reviewer.
- Cut token usage 75-85% by generating Python statistical profiles (P95/Max) instead of passing raw time-series data, lazy-loading exception logs only for approved candidates, offloading all financial math to Python, and implementing local semantic caching - with zero drop in decision quality.

SRE & Operational Automation

- Analyzed production telemetry and executed targeted KQL queries to isolate and eliminate recurring database deadlocks.
- Developed a Python automation utility integrating the Azure CLI and Google Docs API to automate hourly incident logging, eliminating manual tracking overhead while preserving RCA data integrity.

Platform DevSecOps & Secure Infrastructure

- Architected and deployed a zero-trust 3-tier application on Azure, separating the React frontend, backend App Services, and PostgreSQL Flexible Database across isolated Virtual Networks with Private Endpoints to eliminate public internet exposure.
- Implemented system-assigned Managed Identities and Azure Key Vault secret references for passwordless service authentication and zero plaintext credentials in source code.
- Automated infrastructure provisioning and application delivery using modular Azure Bicep templates and Azure DevOps CI/CD pipelines, enabling repeatable deployments across environments.

DevOps Intern **Apr 2025 - May 2025**
HashedIn by Deloitte

- Provisioned and managed cloud resources across AWS and Azure, maintaining environment consistency using Docker, Podman, and Kubernetes.
- Automated CI/CD pipelines with Jenkins and GitHub Actions; wrote reusable Terraform configurations for infrastructure-as-code deployments.

HACKATHON PROJECT

InfraTwin AI - Agentic Cloud OS Prototype **Lyzr AI Hackathon - Top 50 Finalist**
Live Demo

- Built an AI-driven infrastructure design system using LangGraph and React Flow, converting natural-language prompts into visual cloud topologies and JSON blueprints.
- Programmed continuous state drift-detection logic and automated self-healing workflows to simulate recovery during mock resource outages.

PROJECTS

End-to-End 3-Tier Azure Deployment GitHub

- Deployed a full-stack static web app on Azure; integrated Azure DevOps CI/CD to manage the full lifecycle from commit to cloud hosting.

Core DevOps Representation App | Pipeline

- Containerized a Flask & MySQL two-tier app with Docker for environment parity; configured a Jenkins CI/CD pipeline with automated security scanning.

Terrakit - Modular Terraform CLI Automation Tool *Local / In-Development*

- Scripted a modular automation tool to streamline local Terraform workspace setup, enforcing standardized directory structures, state file paths, and boilerplate HCL files.
- Eliminates manual setup overhead and reduces initial workspace configuration friction without relying on heavy external dependencies or LLMs.
- Currently developing as a lightweight local CLI, with plans to package it as a VS Code extension for broader team adoption.

CERTIFICATIONS

- AWS Cloud Architecting - AWS Academy Graduate
- AWS Cloud Foundations - AWS Academy Graduate
- Jenkins for Beginners - Coursera / KodeKloud
- Docker Fundamentals - Coursera / LearnKarts

LANGUAGES

English (Professional) | Hindi (Conversational) | Telugu (Native)

EDUCATION

Bachelor of Technology, Computer Science and Engineering **2021 - 2025**
GITAM University, Bangalore